Project - Low Level Design

on

Multimodal Generative AI system for Hospitality Concept Visualization

Course Name: Generative AI

Institution Name: Medicaps University – Datagami Skill Based Course

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1. Introduction :

The **Multimodal Generative AI System for Hospitality Concept Visualization** is an advanced application that leverages Generative AI to transform textual descriptions into both detailed narratives and high-quality visual representations. The system integrates Large Language Models (LLMs) and text-to-image generation models to provide a comprehensive and interactive way to conceptualize hospitality environments such as hotels, resorts, and restaurants.

This system addresses the growing need for rapid visualization and ideation in the hospitality industry by automating the design interpretation process. Instead of relying solely on manual sketches or expert designers, users can input simple textual prompts and receive complete concept outputs in real time.

1.1. Scope of the document :

This document provides a detailed description of the design, architecture, and implementation of the Multimodal Generative AI system. It covers:

- System architecture and workflow
- Application and component design
- Data and process flow
- API structure and interaction
- Key design considerations

The document is intended to guide developers, stakeholders, and evaluators in understanding how the system operates and how it can be implemented effectively.

1.2. Intended Audience :

This document is intended for:

- Software developers and engineers involved in implementation
- Project evaluators and academic reviewers
- Designers and researchers in Generative AI
- Hospitality professionals exploring AI-based solutions

It assumes a basic understanding of artificial intelligence, APIs, and system design principles.

1.3. System overview :

The system is a multimodal application that processes a single textual input and generates two outputs:

1. A descriptive narrative using an LLM
2. A visual representation using an image generation model The workflow involves:

- Accepting a user prompt describing a hospitality concept
- Processing the prompt in the backend
- Sending the prompt simultaneously to:
  - A text generation model (LLM)
  - An image generation model
- Combining both outputs
- Displaying the final result to the user

This dual-output approach enhances understanding and supports better decision-making in concept design.

2. System Design :

The system follows a modular and scalable architecture designed to support multimodal processing. It integrates frontend, backend, and AI services through APIs to ensure smooth data flow and real-time performance.

2.1. Application Design :

The application is divided into three main layers:

Frontend Layer

- Built using Streamlit or a simple web UI
- Accepts user input (text prompt)
- Displays generated text and image outputs

Backend Layer

- Developed in Python
- Handles request processing and API integration
- Manages communication between frontend and AI models

AI Model Layer

- LLM for text generation
- Image generation model (e.g., Stable Diffusion)

This layered design ensures flexibility and easy integration of different models.

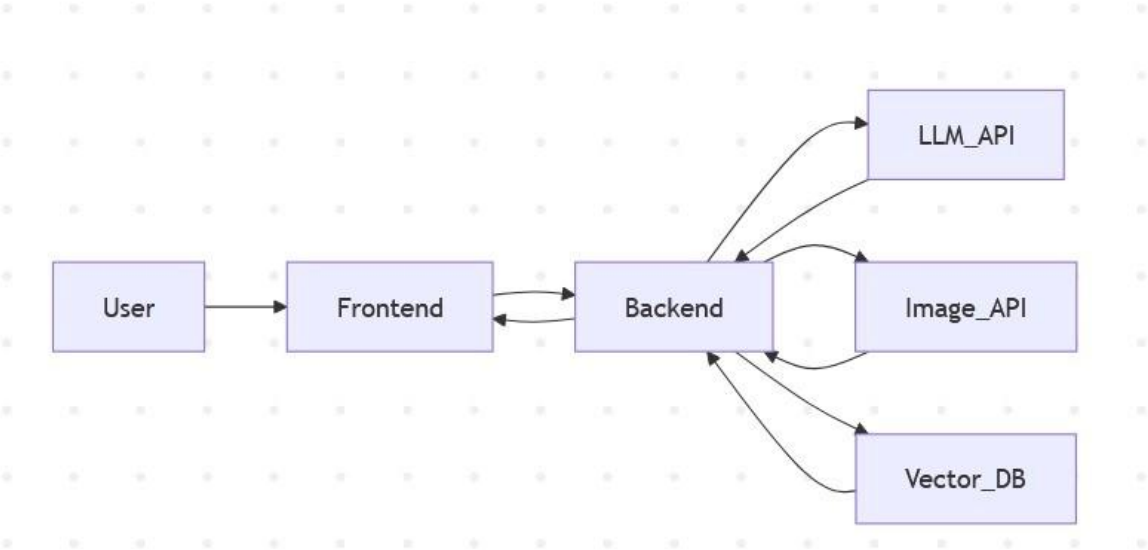

Fig 2.1 : Architectural Diagram

2.2. Process Flow :

The System follows a sequential yet parallel processing flow:

- User enters a textual prompt
- Frontend sends the request to backend API
- Backend validates and processes the input
- The prompt is sent simultaneously to LLM for text generation
- Image model for visual generation
- Both models process the input independently
- Outputs are returned to backend
- Backend combines results
- Final output is displayed on frontend

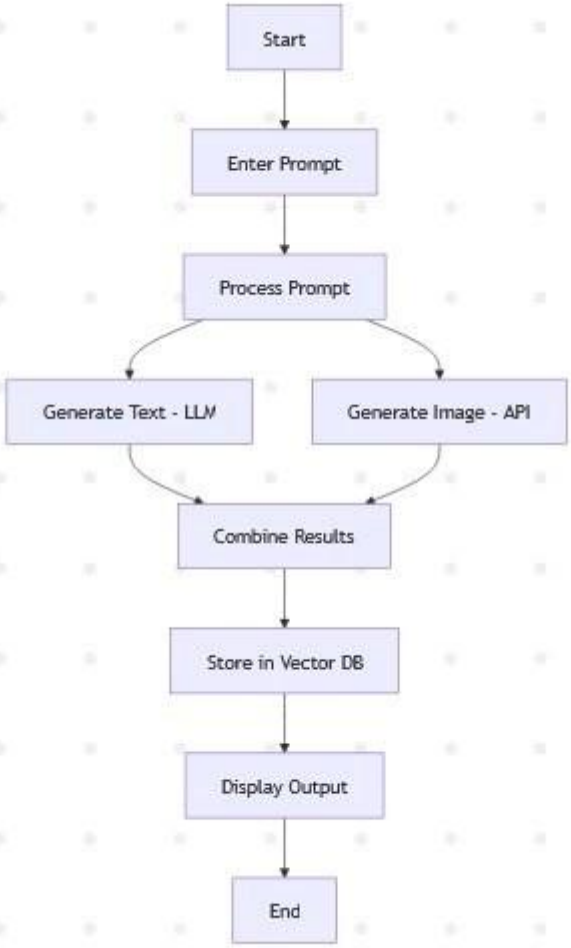

Fig 3.2 : Activity Diagram

2.3. Information Flow :

The information flow in the system is as follows:

- Input Data: User prompt
- Processing Data: Structured prompt passed to AI models
- Output Data:
  - Generated text description
  - Generated image

Flow Direction:

User → Frontend → Backend → AI Models → Backend → Frontend → User

This ensures efficient handling of data and minimal latency.

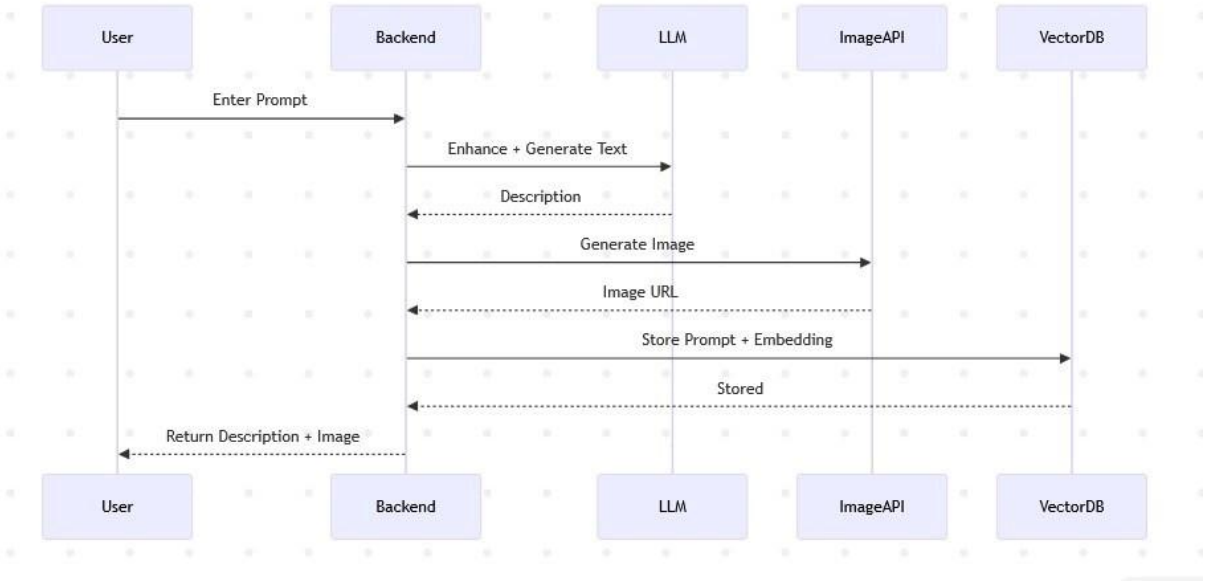

Fig 3.3 : Sequence Diagram

2.4. Components Design :

The system consists of the following components:

1. User Interface

- Input field for prompt
- Output display section

2. API Handler

- Receives requests
- Routes them to appropriate services

3. **Prompt Processor**

- Cleans and formats input text

4. **LLM Module**

- Generates descriptive narrative
- Uses APIs like GPT

5. **Fusion Layer**

- Combines text and image outputs
- Formats response

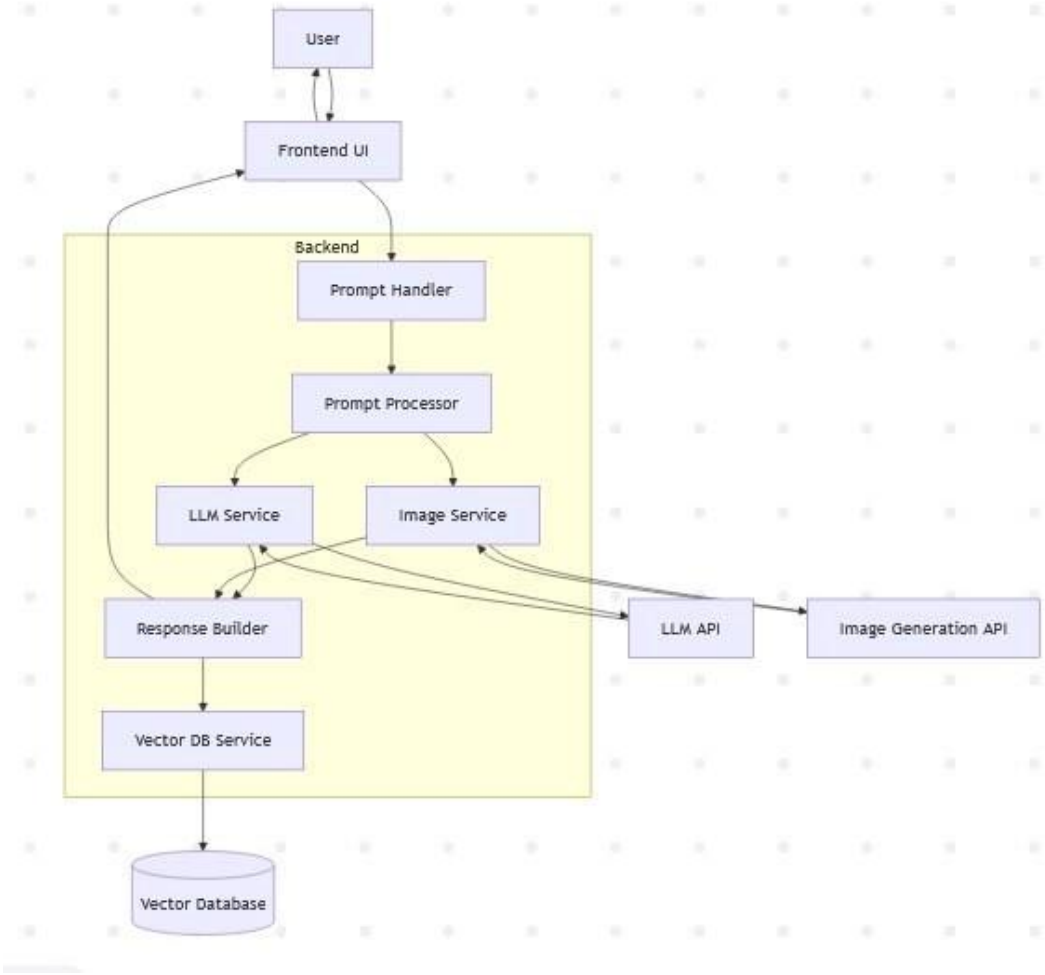

Fig 3.4 : Component Design

2.5. Key Design Considerations :

Several factors were considered during system design:

- **Scalability:** Ability to handle multiple requests
- **Performance:** Fast response time using parallel processing
- **Modularity:** Easy replacement of models
- **Usability:** Simple and intuitive interface
- **Reliability:** Error handling for API failures and invalid inputs
- **Extensibility:** Support for future features like video generation

2.6. API Catalogue :

Method: POST

Request Body:

{

"prompt": "Luxury beach resort with sunset view" }

Processing Steps:

- Validate input
- Send prompt to LLM API
- Send prompt to Image Generation API
- Collect responses
- Merge outputs **Response:**

{
"description": "A luxurious beachfront resort featuring...",
"image_url": "generated_image_link"
}

Error Handling:

- Empty input → return validation error
- API failure → return fallback message
- Timeout → retry or notify user

2.7. Class Diagram

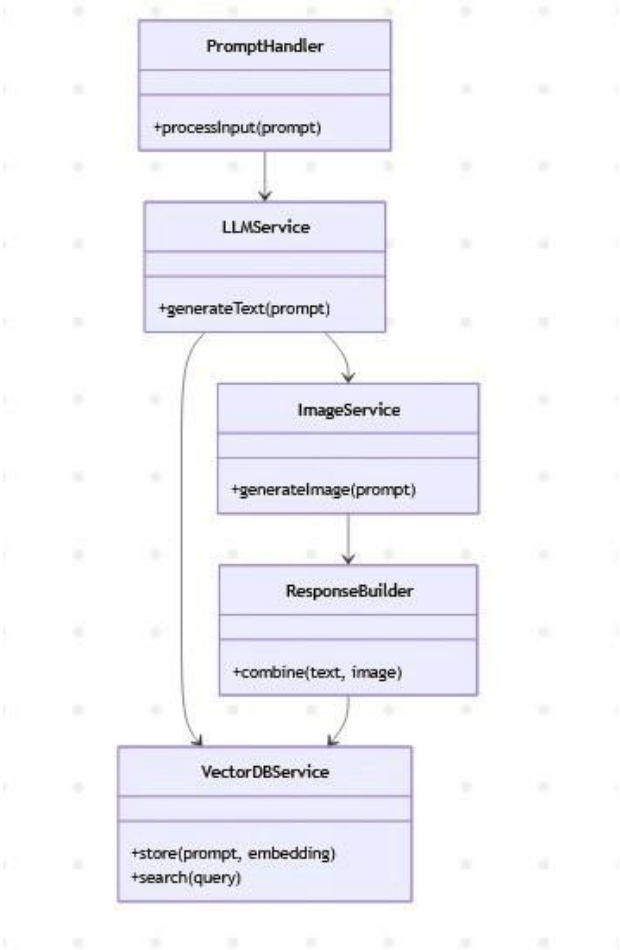

Fig 3.5 : Class Diagram

3. References

- OpenAI GPT API Documentation (for text generation)
- Stable Diffusion Documentation (for image generation)
- Streamlit Documentation (for frontend development)
- Python Official Documentation
- Research papers on Generative AI and Multimodal Systems